

María Fernanda Molina Ron

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PROFILE SUMMARY

Geospatial and data analytics enthusiast with a background in Information Technology and Geoinformatics Engineering. I have hands-on experience in geospatial analysis, WebGIS development, and Python automation, and I've delivered projects ranging from PyQGIS back-end development for QGIS plugins to urban mobility analysis and remote sensing workflows. I also bring strong communication skills from teaching and research. I'm looking for a Data Analyst or GIS/Geospatial Analyst internship where I can apply spatial analytics and data-driven modeling to real-world problems.

EDUCATION

Master in Geoinformatics Engineering

09/2023 – Present

Politecnico di Milano | Milano, Italy

- Relevant Courses: Geospatial Data Analysis, Geospatial Processing, GIS, Space Economy, Earth Observation, Data Mining, Data and Information Quality, Big and Unstructured Data, Machine Learning, Artificial Networks and Deep Learning.

Bachelor of Science in Information Technology Engineering

2014-2020

Yachay University of Experimental Technology and Research (Yachay Tech University) | Imbabura - Ecuador

- Relevant Courses: AI, Neural Networks, Machine Learning, Multivariate Analysis, Databases, Software Engineering. DGPA: 8.7/10

SKILLS

- **Programming**: Python, R, Java, JavaScript
- **WebGIS & Frontend**: React, HTML5, CSS3, OpenLayers
- **Geospatial Development**: PyQGIS (QGIS plugin development), GeoJSON, spatial data validation, rule-based geoprocessing
- **GIS & Spatial Tools**: QGIS, ArcGIS, GRASS GIS, SAGA GIS, GDAL
- **Data Science & ML**: pandas, NumPy, scikit-learn; regression/classification; similarity metrics.
- **Remote Sensing & Raster Analytics**: Sentinel-2, Landsat 8; NDVI/NDRE/NDWI; raster preprocessing and change detection
- **Core Competencies**: Data Analysis & Machine Learning, GIS & Spatial Analytics, WebGIS Applications, Remote Sensing (multispectral indices and change detection), Risk/Susceptibility & Exposure Mapping.

EXPERIENCE HIGHLIGHT

Geospatial Software Developer (QGIS Plugins) — External Consultant

10/2025 – Present

Public-sector GIS modernization project (NDA) | Colombia

- Developed core back-end algorithms for a modular QGIS plugin suite for municipal road network management and addressing.
- Applied Python/PyQGIS geoprocessing, spatial data validation, and rule-based logic to transform cadastral and orthophoto layers into standardized road-network and addressing outputs.

Note: Code and client materials are confidential; repository is private under NDA.

Teacher (Coding / Project Design, Math & Physics)

12/2020 – 04/2023

SEK Ecuador International School | Quito - Ecuador

- Taught Design Cycle methodology to over 140 students, emphasizing project management and design principles.
- Mentored students in web programming, game design, environmental campaigns, and social initiatives.
- Delivered high-school Math and Physics instruction; prepared students for the International Baccalaureate (IB) Physics exam.

Product Designer and Project Leader

01/2022 – 06/2022

CYORI S.A.S (Quito - Ecuador) | Quito - Ecuador

- Led a team to develop UProfessor, Ecuadorian web application focused on education, completing a functional prototype within 6 months and obtain financing proposal from national incubator Prendho
- Product Designer of UProfessor's website. UProfessor,
- Figma Project: <https://www.figma.com/file/naYN9fyNoDMJRjsbDVh3kJ/UProfessor?node-id=0%3A1>

PROJECTS

Active Mobility Data Analysis: Walkability & Bikeability Similarity Across Capital Cities

06/2025

[View Project Repository](#)

- Conducted walkability/bikeability analysis across 20 mid-sized capital cities using 11 street-network metrics.
- Computed and evaluated city similarity with Mahalanobis distance to account for correlated indicators.
- Identified key drivers and non-obvious pairings (connectivity/street length; circuitry; terrain steepness).

Tools & Technologies: Python, Pandas, NumPy, scikit-learn, Jupyter/Colab, GIS data processing.

Urban Network Similarity Analyzer (WebGIS): Dual-Map City Comparison Tool

06/2025

[View Project Repository](#)

- Built a WebGIS application for side-by-side city comparison in Walkability and Bikeability modes.
- Implemented Top-3 similarity ranking using precomputed Mahalanobis distance matrices with metric-table interpretability.
- Delivered a responsive dual-map UI with interactive selection and ranking updates.

Tools & Technologies: React, OpenLayers, Django REST API, Python, GeoJSON.

Insurance Proposal: Avalanche Susceptibility and Exposure Assessment Using Dynamic Meteorological Data

05/2025

[View Project Slides](#)

- Developed a proposal for Unipol to improve underwriting with dynamic avalanche risk maps using Earth Observation and climate reanalysis data.
- Designed a workflow combining terrain and ERA5-downscaled weather variables to update risk surfaces over time.
- Produced an exposure approach linking risk maps to buildings/assets to support pricing and risk selection.

Tools & Technologies: GIS, ERA5/downscaled data, supervised ML, raster processing, exposure analytics

Flood Detection and Damage Assessment in St. Peter, Minnesota Using Landsat 8 Imagery

08/2024

[View Project Repository](#)

- Conducted remote sensing using Landsat 8 imagery to assess flood impact and damage to buildings and roads.
- Developed a workflow for cloud masking, pansharpening, and NDWI calculation to detect changes in water coverage.
- Generated change detection maps and performed proximity analysis to evaluate infrastructure risks.

Tools & Technologies: QGIS, Python (Google Colab), Landsat 8, OpenStreetMap, NumPy, GDAL, Rasterio.

NDVI and NDRE Vegetation Health Analysis near Ilva Steel Plant, Italy (2018-2024)

08/2024

[View Project's Results](#)

Evaluated vegetation health near the Ilva Steel Plant to assess environmental reforms.

- Developed a workflow for multispectral image processing and supervised classification to monitor vegetation health changes.
- Utilized NDVI and NDRE indices to assess vegetation stress.

Tools & Technologies: QGIS, Sentinel-2, CORINE Land Cover, NDVI, NDRE, Dzetsaka.

RESEARCH & RELEVANT AWARDS

GenHackaton 2025 - Winner of "Most Meaningful Visualisations" Prize

12/2025

Project: Measuring the Urban Heat Island Effect

Repository: <https://github.com/FernandaMolina22/genhack2025-era5-uhi-team36> (certificate pending).

Undergraduate Research Thesis

07/2020

"Comparison of radial distortion correction models for self photogrammetric camera calibration."

Available from: <https://repositorio.yachaytech.edu.ec/handle/123456789/218?locale=en>

Conference

05/2019

"Red-Black Tree Based NeuroEvolution of Augmenting Topologies." - William R. Arellano, Paul A. Silva, Maria F. Molina, Saulo Ronquillo, Francisco Ortega-Zamorano. International Work-Conference on Artificial Neural Networks. IWANN 2019: Advances in Computational Intelligence pp 678-686. [DOI:10.1007/978-3-030-20518-8_56](https://doi.org/10.1007/978-3-030-20518-8_56).

LANGUAGE PROFICIENCY

- Spanish (native speaker)
- English (B2 - TOEFL iBT score (12/2020) = 84)
- Italian: A2/B1 (self-assessed)